# Fraud Risk Command Center

End-to-end analytics + ML fraud pipeline with SQL warehouse + Power BI command center

Prepared on: 27 Feb 2026

## 1. Executive Summary

- Built a production-style fraud analytics pipeline: raw → staging → curated dimensions → marts fact, enabling governed reporting and investigation workflows.
- Processed **5,000** transactions with **23.42%** fraud rate; created a command-center dashboard with executive, operations, driver intelligence, and monitoring views.
- Engineered behavior + velocity features (e.g., txn_count_1h/24h, avg_amount_7d, max_amount_24h, merchant_fraud_rate) and trained baseline + strong models.
- Selected an operating threshold (~**0.59**) using F1 tuning; produced risk bands (Low/Medium/High) and top driver “reason codes” to support investigations.
- Delivered portfolio-ready assets: SQL scripts, notebooks, model scoring output, and Power BI dashboards suitable for Big4-style storytelling.

## 2. Business Problem

Retail payments generate large transaction volumes with meaningful fraud exposure. The goal is to detect and prioritize likely-fraud transactions, understand fraud patterns (where/when/how), and operationalize investigation via an analyst workbench and monitoring controls.

## 3. Data & Entities

- Source: Enterprise Fraud Transaction Dataset (Excel) → SQL Server ingestion.
- Grain: 1 row = 1 transaction (transaction_id).
- Key fields: customer_id, merchant_id, merchant_category, transaction_timestamp, transaction_amount, channel, payment_method, device_id/device_type, customer_region, is_international, customer_age, account_age_days, previous_fraud_flag, fraud_label.
- Size used in build: 5,000 rows.

## 4. Architecture (Production-style)

- **staging**: raw ingested transactions (stg_transactions).
- **curated**: conformed dimensions (Dim_Customer, Dim_Geo, Dim_Merchant, Dim_Device, Dim_Channel, Dim_Time).
- **marts**: analytics-ready fact (Fact_Transactions) + scored/ops queue tables for Power BI.
- Governance fields on tables: ingestion_dt, source_system, load_batch_id, record_hash.

## 5. Data Engineering & Quality (Phase 3)

- Validated key quality rules: invalid timestamps, negative/zero amounts, impossible ages, duplicate transaction IDs.
- Applied cleaning policy (documented): coerce timestamps, filter invalid rows, constrain ages to plausible range, de-duplicate by transaction_id (keep latest).

- Exported versioned clean file: clean_transactions_v1.csv.

| Metric | Value |
|---|---|
| Raw rows | 5000 |
| Invalid timestamp rows | 0 |
| Invalid amount rows | 0 |
| Duplicate transaction_id rows | 0 |
| Clean rows | 5000 |

## 6. Feature Engineering (Phase 5)

- **Customer behavior**: txn_count_1h, txn_count_24h, avg_amount_7d, max_amount_24h, time_since_last_txn.
- **Transaction context**: hour/day/weekend flags; channel/payment method encodings.
- **Merchant risk**: merchant_fraud_rate (historical fraud rate by merchant/category).
- **Geo & device**: geo_switch_flag / device switch indicators (where available) + international flag.

## 7. Modeling & Scoring (Phase 6)

- Baseline: Logistic Regression (balanced class weights) with scaling to improve convergence.
- Strong model: Random Forest; compared ROC-AUC and PR-AUC (fraud is imbalanced).
- Threshold tuning: selected threshold around 0.59 to optimize F1; produced risk bands and reason codes.

| Metric | Logistic Regression (scaled) | Random Forest |
|---|---|---|
| ROC-AUC | 0.7256 | 0.7002 |
| PR-AUC | 0.5685 | 0.5588 |
| Chosen threshold | 0.59 (F1-optimized) | 0.59 (applied to probability output) |

**Top risk drivers (Random Forest feature importance)**

- 1. previous_fraud_flag
- 2. avg_amount_7d
- 3. max_amount_24h
- 4. account_age_days
- 5. transaction_amount
- 6. is_international
- 7. customer_age
- 8. txn_hour
- 9. merchant_fraud_rate

## 8. Fraud Analytics & Investigation Workbench (Phase 4)

- Trend monitoring: daily/weekly fraud rate and fraud loss trends.
- Velocity checks: multiple transactions per customer in short time windows.
- Amount anomaly: baseline vs current amount using rolling averages.
- Geo/device switching: identify unusual changes for a customer.
- Merchant/category risk: fraud concentration by merchant category and channel.
- Cohorts: fraud rate by customer signup cohort (where available).
- Rules engine: triggered-rule table per transaction for explainable flags.

## 9. Power BI Dashboard (Phase 7)

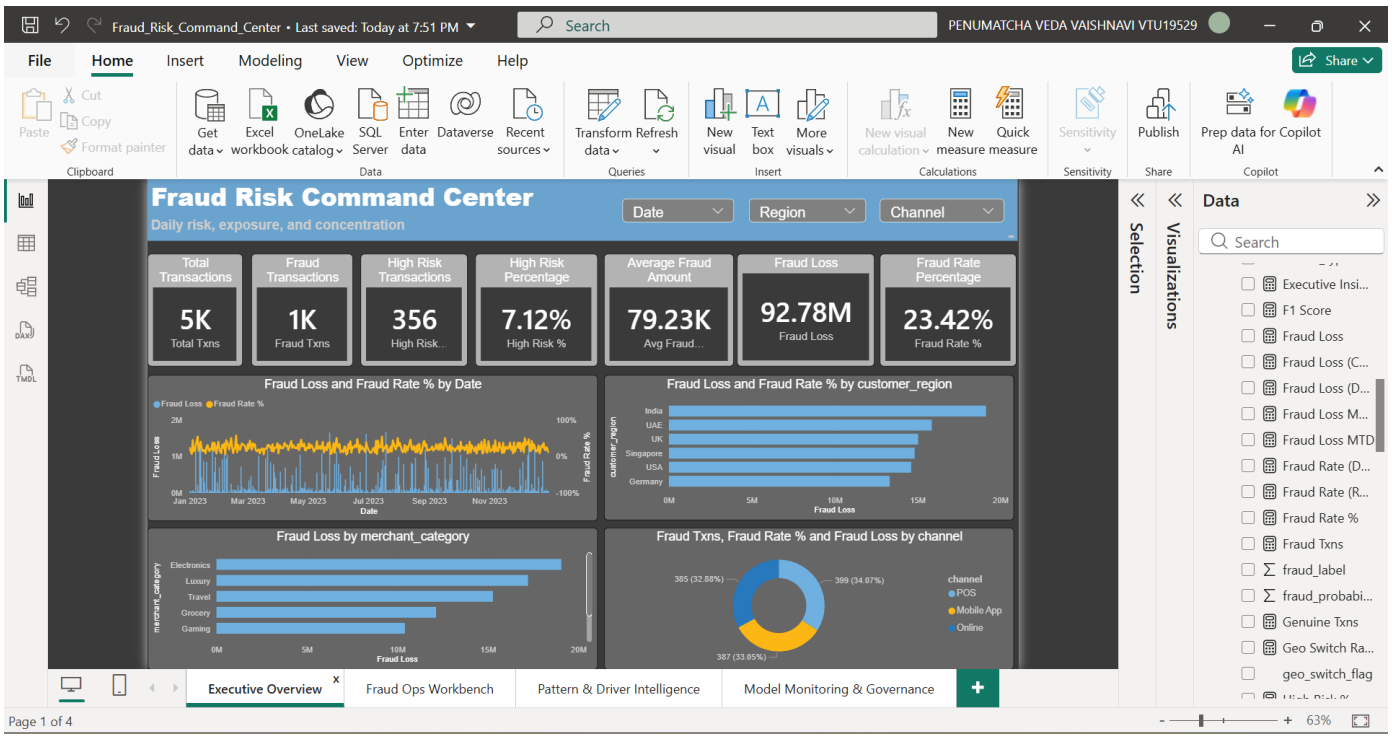
- **Dashboard 1 — Executive Overview**: fraud loss, fraud rate, high-risk % and concentration across region/category/channel; time trends.
- **Dashboard 2 — Fraud Ops Workbench**: investigation queue with sortable high-risk list, SLA-style queue volumes by risk tier, and drill-down.
- **Dashboard 3 — Pattern & Driver Intelligence**: heatmap (category × channel fraud rate), geo/device switching views, anomaly scatter, feature importance.
- **Dashboard 4 — Model Monitoring & Governance**: ROC/PR cards, threshold impact (precision/recall/F1), risk band calibration, drift proxy.

## 10. Recommendations (Phase 8 preview)

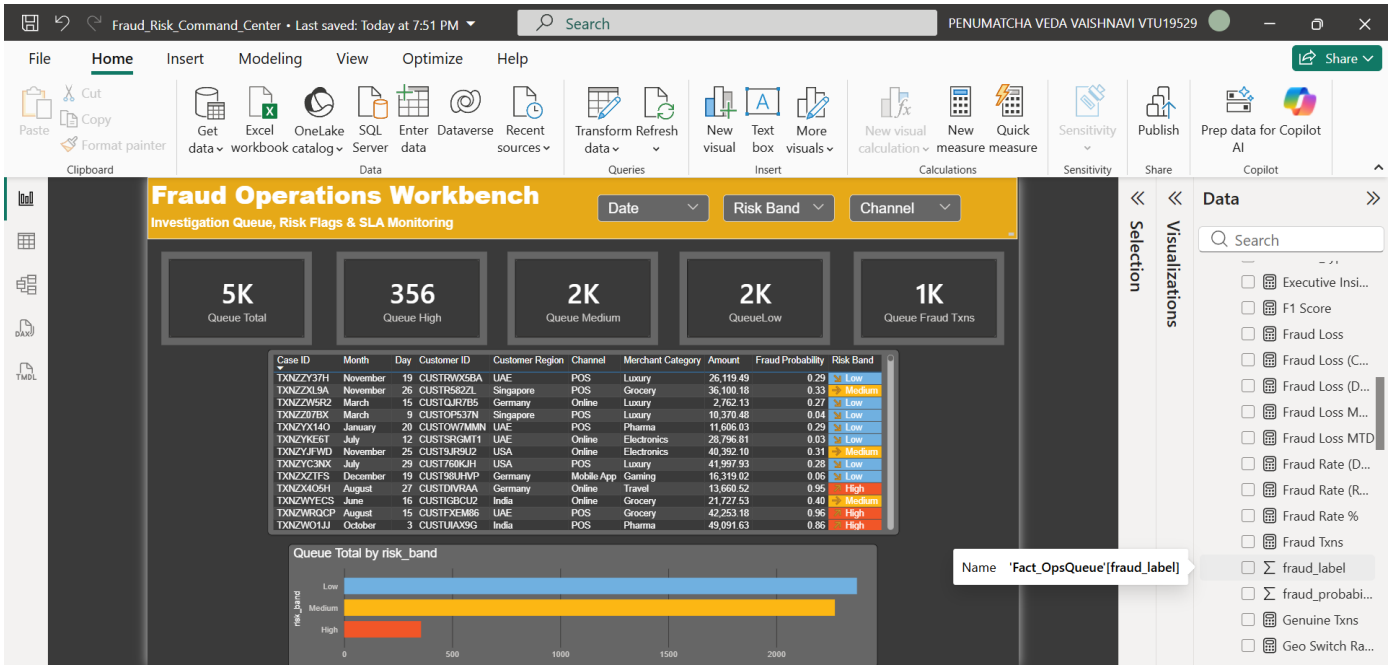
- Prioritize **High** risk band for manual review; use Medium for rule-based secondary checks; Low for monitoring only.
- Implement alerts for spikes in fraud loss, geo switch rate, and international transaction concentration.
- Add rules: previous_fraud_flag + international + high amount; abnormal amount vs avg_amount_7d; rapid velocity within 10–60 minutes.
- Operationalize weekly model monitoring: PR-AUC, threshold impact, and drift proxy (avg fraud_probability over time).

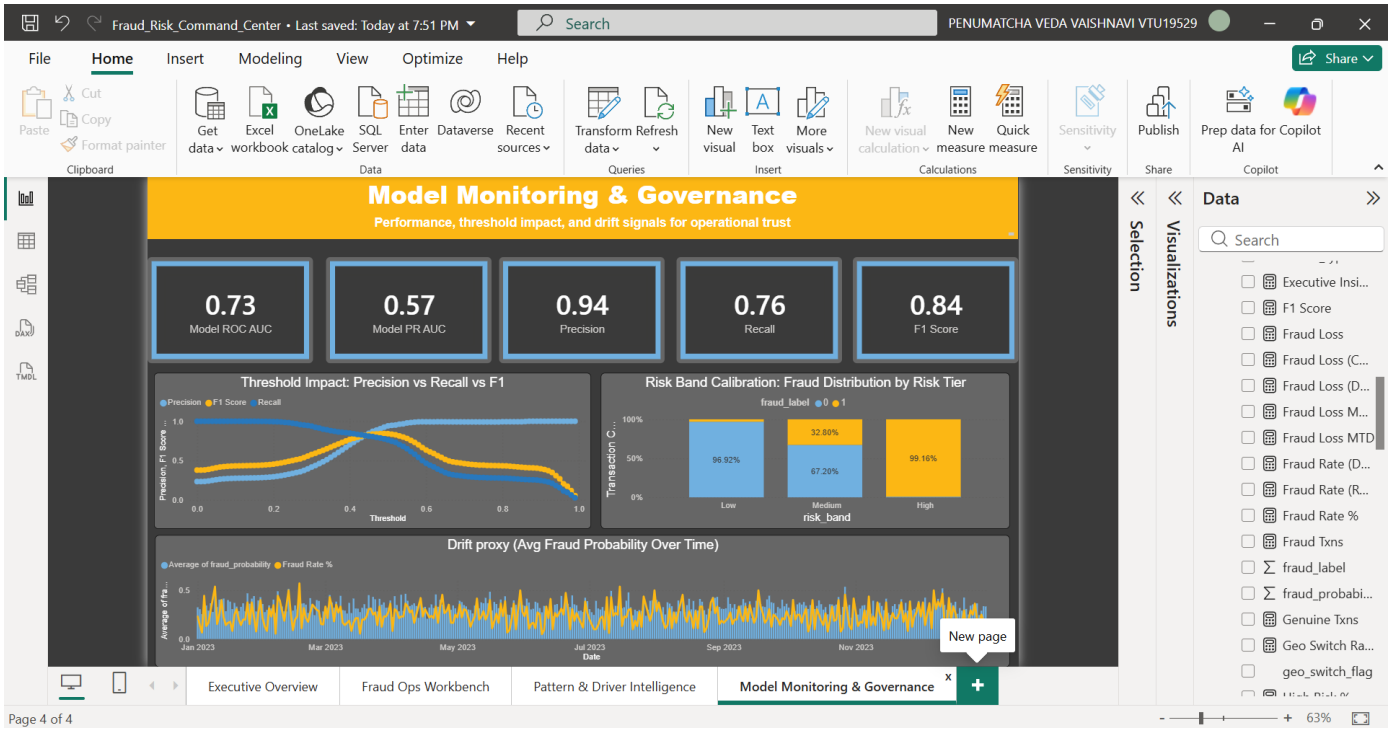
# Appendix A — Dashboard Screenshots

Dashboard 1 screenshot




Dashboard 2 screenshot




Dashboard 3 screenshot

Dashboard 4 screenshot